VIVA QUESTIONS WITH ANSWERS

1. What is Artificial Intelligence (AI)?

   Answer: Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think and act like humans.

2. What is an intelligent agent?

   Answer: An intelligent agent is a system that perceives its environment and takes actions to achieve its goals autonomously.

3. What are the types of environments an agent can operate in?

   Answer: Agents can operate in environments that are deterministic, observable, static, and dynamic, depending on the properties of the environment.

4. What is the difference between a reflex agent and a goal-based agent?

   Answer: A reflex agent acts based on current perceptions using predefined rules, while a goal-based agent evaluates actions to achieve specific goals.

5. What is a utility-based agent?

   Answer: A utility-based agent selects actions that maximize its expected utility or satisfaction based on available options.

6. What is the state space in AI?

   Answer: The state space is the set of all possible states that can be reached during the process of solving a problem.

7. What are the components of a production system?

   Answer: A production system consists of rules, a knowledge base, an inference engine, and working memory.

8. What is the difference between a well-defined problem and an ill-defined problem?

   Answer: A well-defined problem has clear objectives and constraints, while an ill-defined problem lacks clear goals and solutions.

9. What are the key issues in designing search programs?

   Answer: Key issues include time complexity, space complexity, optimality, completeness, and efficiency in solving a problem.

10. What is the role of fuzziness in fuzzy logic systems?

    Answer: Fuzziness in fuzzy logic allows for reasoning and decision-making with imprecise, uncertain, or approximate information.

11. What is a problem-solving agent?

    Answer: A problem-solving agent is an intelligent agent that takes actions to reach a desired goal from an initial state through a search process.

12. What is the Breadth-First Search (BFS) strategy?

    Answer: BFS is a search strategy that explores all possible nodes at the present depth level before moving on to nodes at the next depth level.

13. What is Depth-First Search (DFS)?

    Answer: DFS is a search strategy that explores as far down a branch of the tree as possible before backtracking to explore other branches.

14. What is Depth-Limited Search?

    Answer: Depth-Limited Search is a variation of DFS that limits the depth of search to a pre-defined depth level to prevent infinite loops.

15. What is Bidirectional Search?

    Answer: Bidirectional Search simultaneously searches from both the initial state and the goal state, meeting in the middle to find the solution more efficiently.

16. What is Greedy Best-First Search?

    Answer: Greedy Best-First Search selects the node that appears to be closest to the goal based on a heuristic function.

17. What is the A Search algorithm?*

    Answer: A* Search is a heuristic search algorithm that evaluates nodes based on both the cost to reach the node and the estimated cost to the goal, providing optimal solutions.

18. What is AO Search?*

    Answer: AO* Search is an algorithm used for solving problems in a Directed Acyclic Graph (DAG) by combining the concepts of A* and AND/OR graphs.

19. What is Hill Climbing Search?

    Answer: Hill Climbing Search is a local search algorithm that continuously moves towards the neighbor with the highest value, optimizing a specific objective function.

20. What is Simulated Annealing?

    Answer: Simulated Annealing is an optimization algorithm that explores the solution space by allowing occasional moves to worse solutions to escape local optima, inspired by the annealing process in metallurgy.

21. What is local search for constraint satisfaction problems (CSPs)?

    Answer: Local search for CSPs involves iteratively improving a candidate solution by making small changes to reduce constraint violations, aiming to find a solution that satisfies all constraints.

22. What is adversarial search in AI?

    Answer: Adversarial search is a process where agents compete against each other, and each agent aims to maximize its own benefit while minimizing the opponent's benefit, typically used in games like chess.

23. How are optimal decisions made in games?

    Answer: Optimal decisions in games are made by evaluating all possible moves and selecting the one that maximizes a player's chances of winning, considering both the current state and possible future moves.

24. What is the Min-Max search procedure?

    Answer: The Min-Max search procedure is an algorithm used in game theory to recursively determine the best move by assuming that the opponent will also play optimally, minimizing the maximizing player’s benefit.

25. What is Alpha-Beta pruning in adversarial search?

    Answer: Alpha-Beta pruning is an optimization technique for Min-Max search that eliminates branches of the game tree that will not affect the final decision, improving search efficiency.

26. . What is knowledge representation in AI?

    Answer: Knowledge representation in AI is the process of encoding information about the world in a form that a computer system can use to solve complex tasks like reasoning and problem-solving.

27. What is rule-based knowledge representation?

    Answer: Rule-based knowledge representation uses a set of "if-then" rules to encode knowledge, where each rule represents a relationship or condition about the world.

28. What is the difference between procedural and declarative knowledge?

    Answer: Procedural knowledge describes how to perform tasks or actions, while declarative knowledge represents facts or information about the world.

29. What is logic programming in AI?

    Answer: Logic programming is a programming paradigm where programs are written as sets of logical statements, and computation is performed through logical inference.

30. What is the difference between forward and backward reasoning?

    Answer: Forward reasoning starts from known facts and applies rules to reach a conclusion, while backward reasoning starts from a goal and works backward to find supporting facts.

31. What is Bayesian Probability and Bayes' Theorem?

    Answer: Bayesian Probability uses Bayes' Theorem to update the probability of a hypothesis based on new evidence, providing a method to calculate conditional probabilities.

32. What are Certainty Factors in Rule-Based Systems?

    Answer: Certainty factors quantify the degree of belief or confidence in a conclusion drawn from a rule-based system, allowing reasoning under uncertainty.

33. What are Bayesian Networks?

    Answer: Bayesian Networks are graphical models that represent probabilistic relationships between variables, where nodes represent variables and edges represent conditional dependencies.

34. What is the Dempster-Shafer Theory?

    Answer: The Dempster-Shafer Theory is a mathematical framework for reasoning with uncertainty, allowing the combination of evidence from different sources to assign belief functions to hypotheses.

35. What is a Markov Decision Process (MDP)?

    Answer: A Markov Decision Process is a mathematical model for decision-making where an agent interacts with an environment in discrete time steps, with outcomes determined by both the agent's actions and probabilistic transitions.